Task 02 – Reflection Essay

Going into this task I assumed the deep-learning part would be the mountain and the geometry a footnote. It turned out the other way round. Running YOLOv8 on the avenue photo and getting a labelled bicycle box back was a single function call. What actually took thought was believing – and then proving to myself – that a flat photograph with no depth information can still yield a real distance in metres.

The hinge of the whole task is the cross ratio. Perspective destroys ordinary measurement: the avenue's two edges visibly converge towards the arch, so equal real lengths shrink as they recede, and "measure pixels, multiply by a constant" simply does not work. The cross ratio is the one combination of four collinear points that a perspective camera leaves unchanged, so its value in the photo equals its value on the ground. That single invariant is what lets three known distances pin down a fourth unknown one. Seeing the link back to the 25-May surveillance-camera example – locating a train between two stations using the point where the rails meet – was the moment the idea clicked.

Two practical lessons stuck with me. First, the arch and the vanishing point are essentially the same point in this image, so I could not use both as references; their pixel positions were too close and the arithmetic went unstable. Moving to the intersection as the middle reference cured it. Second, and more striking, the choice of point inside the YOLO box mattered enormously. The box is a rectangle but the bicycle only meets the ground at the bottom; using the box centre instead of the wheel contact moved the answer by more than twenty metres. Near the vanishing point one pixel is worth roughly 0.16 m, so where you "stand" the object is not a detail – it is the dominant source of error. By contrast, rounding my pixel readings to the nearest ten changed the result by under a metre. The modelling assumption, not the measurement precision, was the thing to get right.

I cross-checked the bicycle distance two independent ways, once treating the vanishing point as the point at infinity and once with three finite landmarks, and the two agreed to within a few metres. That agreement did more for my confidence than any single calculation could have.

If I were to redo it, I would spend longer nailing the vanishing point, since real path edges are not perfectly straight and that uncertainty propagates into everything downstream. The broader takeaway is that the machine-learning tool is a convenience that locates an object; the understanding lives entirely in the projective geometry and in respecting the assumptions – collinearity, a level camera, an object truly on the line – that the cross ratio quietly depends on.